

Austin Peay State University

DC/AC CIRCUITS

Lab 1. Basic Components and Devices.

Objective:

In this laboratory you will learn about the main characteristics of circuit components, how to use the multimeter to determine resistance and capacitance, quick check of diodes for failures; will measure I-V characteristic of various devices

Parts:

- Resistors, capacitors, diodes, bulbs

Equipment:

Digital multimeter (DMM), Variable power supply, Voltmeter, Ammeter, Access to Internet

Instrumentation

The multimeter you'll be using in this lab is the **Mastech My64** DMM, shown in Figure. A multimeter is a device used to take measurements of electrical quantities such as resistance, current, voltage and capacitance. DMM stands for **Digital Multi Meter**.

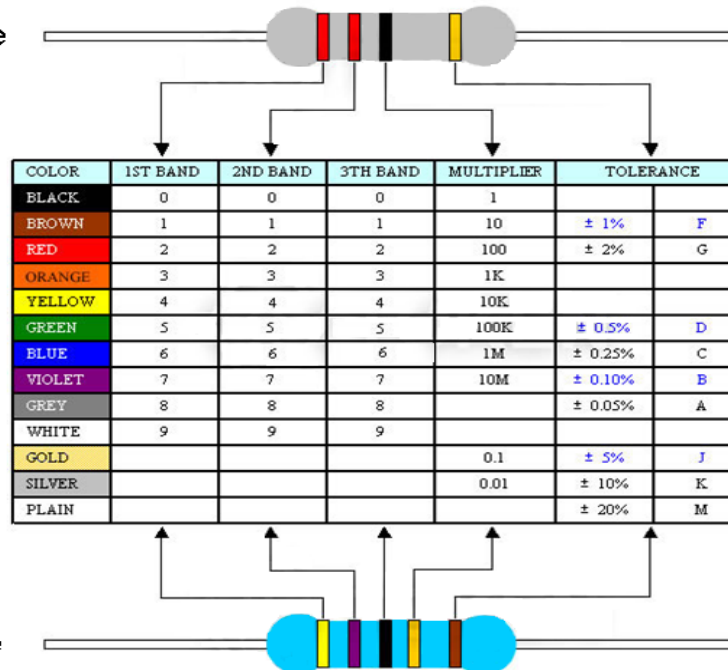
Caution: Never connect an ammeter directly across a power supply as it will cause a short circuit and will certainly damage the meter.



Part A - Measurements:

Resistance measurements

4-band code



5-band code

How to read
the resistor code

First find the tolerance band, it will typically be gold (5%) and sometimes silver (10%). Starting from the other end, identify the first band - write down the number associated with that color; in this case Red is 2. Now 'read' the next color, here it is red so write down a 2 next to the two. (you should have '22' so far.) Now read the third or 'multiplier' band and write down that number of 1. In this example, the 'multiplier' band is Black so we get 22 Ω.

If the 'multiplier' band is Gold move the decimal point one to the left. If the 'multiplier' band is Silver move the decimal point two places to the left.

Select 5 different resistors (write down their color coded values).

- Draw the circuit symbol for the resistor
- Determine the nominal value as indicated by the resistor color code.
- Set the multimeter to measure resistance. Determine the actual values of the resistance.

Comparison:

Tabulate the nominal value and the measured value of the set of resistors.

Compute and tabulate the percent difference between the actual (measured) value of the resistors and their nominal value as indicated by its color code.

Data

Characteristics	Resistor Number				
	1	2	3	4	5
1 st Color					
2 nd Color					
3 rd Color					
4 th Color					
5 th Color					
Coded Value					
Coded Tolerance %					
Measured Value					

Capacitance measurements:**Select 5 different capacitors.**

- Set the multimeter to measure capacitance.
- Determine the actual values of the capacitance and compare to the nominal ones.

Data

Characteristics	Capacitor Number				
	1	2	3	4	5
Type (fixed, variable, electrolytic, etc.)					
Capacitance measured					
Capacitance nominal					

Select 5 (minimum 3) different diodes.

- Set the multimeter to measure resistance (choose kOhm range). Perform a quick failure check for each diode (see published in D2L notes for Chapter 1-2 on how to do failure check).
- Find in the internet datasheets for your particular diodes. Which are the other characteristics of the diodes? Put them into table below.

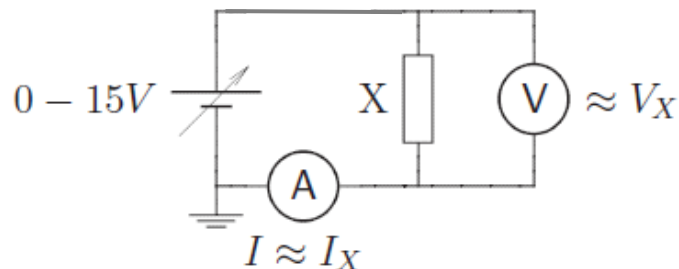
Data

Characteristics	Diode Number				
	1	2	3	4	5
Label (make)					
Draw diode and label anode and cathode terminals on the picture					
Resistance in forward bias					
Resistance in reverse bias					
Status (Ok or failure)					
Maximum current					
Frequency range					

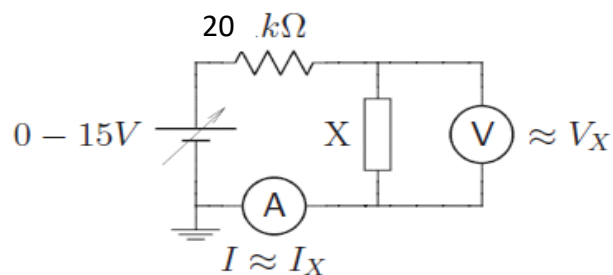
Part B – I-V characteristics

In this activity, you will learn how to obtain volt-ampere (I-V) characteristics of passive circuit elements. A passive element is a two-contact device that contains no source of power or energy; an element that has a power source is called an “active” element.

- 1) Select the **resistor** with resistance between $20\text{ k}\Omega$ and $50\text{ k}\Omega$.
- 2) Build a circuit below, connecting your chosen resistor to the place marked X on the diagram.



- 3) Connect the circuit to +15 V terminal of variable power supply. Start from 0 V and gradually increase the voltage (in 0.5 V steps), recording simultaneously voltage V_x and current I_x .
- 4) Plot these data as $I=f(V)$ and attach the plot to your report.
- 5) Discuss the observed I-V characteristic. What does the slope of the curve mean?
- 6) Replace the element X with a 12 V **light bulb**.
- 7) Repeat steps 3 through 5.
- 8) Replace the element X in the circuit below with a **diode**. Use DMM to measure V_x .



- 9) Connect the circuit to +15 V terminal of variable power supply. Start from 0 V and gradually increase the voltage (in $\sim 0.05\text{ V}$ steps), recording simultaneously voltage V_x and current I_x .
- 10) Connect now the circuit to -15 V terminal of variable power supply (or simply switch the + and – terminals if only +15 V is available). Start from 0 V and gradually change the voltage (in 0.5-1.0 V steps), recording simultaneously voltage V_x and current I_x .
- 11) Repeat steps 4 and 5.